BAYESIAN NETWORK CLASSIFICATION

A SIMPLE EXAMPLE: Classification by Probabilistic Inference

(the Cancer network)

BAYESIAN NETWORK CLASSIFICATION

A Network of Causes

Goal: represent how variables depend on one another, then answer questions by **computing probabilities**. This works in any direction.

THE GRAPH

Which cause affects what

A directed acyclic graph. Each node is a variable; each arrow encodes a direct dependence, usually read as cause → effect.

“the blueprint”

THE TABLES

How strong each link is

For discrete variables, every node carries a Conditional Probability Table (CPT): the probability of that variable given the values of its parents.

“the measurements”

Why use it?

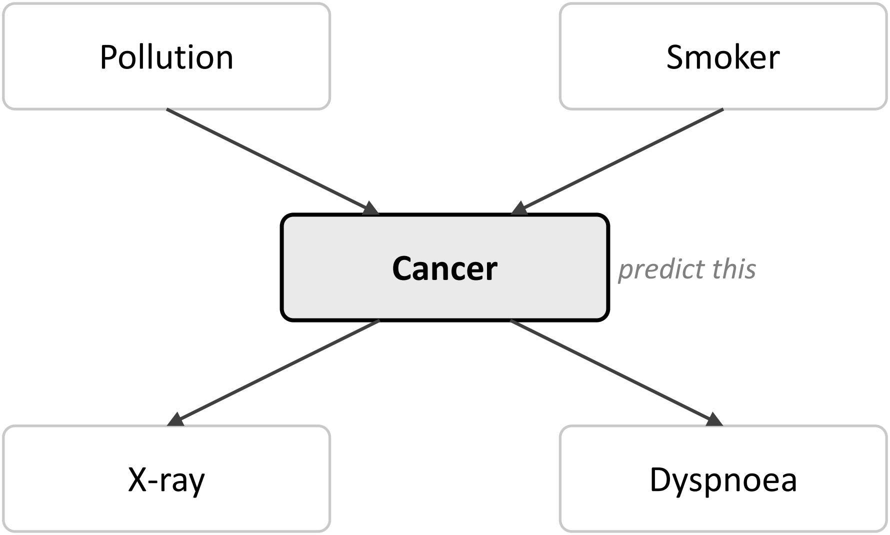
- Transparent & auditable: you can read off why a prediction was made.
- Encodes expert knowledge and copes with small or partially missing data.
- Returns full probabilities, not just a label, and reasons in any direction (effects → causes, or causes → effects).

A SIMPLE EXAMPLE:

Diagnosing from Evidence

Observe what we can see (**evidence**), then let the network compute the probability of the hidden class we care about (here, the Cancer node).

The network



Classification = one inference

Observe **Smoker = yes, Dyspnoea = present**.
Ask the network: **P(Cancer | evidence)?**
Evidence propagates through the CPTs, returning a probability for each class.

Belief updates as evidence arrives

Start from the prior (cancer is rare). Add evidence one piece at a time and watch the posterior **P(Cancer)** move. The most probable state is the predicted class.

FROM TOY PROBLEM TO INDUSTRY

Where Bayesian Networks Lead

The same observe → infer recipe is used where decisions must be explainable and data is limited.

MEDICINE

Clinical decision support

Observe

symptoms and test results

Infer

probability of each disease

Diagnostic aids where clinicians need to see the reasoning.

RELIABILITY & SAFETY

Fault & risk diagnosis

Observe

sensor readings and alarms

Infer

the most likely failure cause

Aerospace, energy, manufacturing; pairs with fault-tree analysis.

FINANCE & OPERATIONS

Risk & fraud scoring

Observe

transaction and profile signals

Infer

probability of fraud or default

Used where each decision must be auditable, not a black box.

Reality check:

- Naive Bayes — the maximally simplified Bayesian network — is still a useful ultra-lightweight baseline for text classification and the like. However, this is a special case.
- Full Bayesian network classifiers themselves are rarely chosen in modern ML practice: the modeling effort is high and the accuracy seldom beats gradient boosting or deep models.
- That said, the same framework still serves as the backbone of causal inference and probabilistic programming. Even in domains where explainability matters most, **much of the action has shifted from Bayesian-network classifiers to causal models**.
- Judea Pearl, who first formalized Bayesian networks, made the same turn himself, going on to develop the structural causal model (SCM) and the modern theory of causal inference.

SUPPLEMENT

“Bayesian” Methods at a Glance

Method	Type	What it gives you	Typical use
GROUP 1 · Graphical models			
Naive Bayes	Simple generative classifier	A fast class label	Text / spam baseline
Bayesian network	**Directed probabilistic graphical model**	**Probabilistic reasoning over variables**	**Diagnosis, risk modeling**
GROUP 2 · Bayesian predictive models			
Bayesian linear regression	Regression with uncertainty	A predictive distribution	A/B testing, marketing mix
Bayesian neural network	Neural network with Bayesian weight uncertainty	Predictions with uncertainty	Safety-critical ML, OOD awareness
GROUP 3 · Optimization procedures			
Bayesian optimization	Optimization procedure using a probabilistic surrogate model	The next best point to try	Hyperparameter / experiment tuning

Bottom line: Naive Bayes is the simplest special case of a Bayesian network. A Bayesian neural network is a neural network with Bayesian uncertainty, not a Bayesian network.

APPENDIX 1:

Risk Factors -> Cancer

- Pollution and Smoking together -> Cancer

Smoker	Pollution	P(Cancer=True)	P(Cancer=False)
smoker	low	0.030	0.970
	high	0.050	0.950
non_smoker	low	0.001	0.999
	high	0.020	0.980

Base rates that seed the network (no incoming arrows).

Smoker	P
smoker	0.3
non_smoker	0.7

Pollution	P
low	0.9
high	0.1

APPENDIX 2:

Cancer -> Symptoms

- Cancer -> Xray

Cancer	P(Xray=True)	P(Xray=False)
True	0.90	0.10
False	0.20	0.80

True = positive, False=negative

- Cancer -> Dyspnoea

Cancer	P(Dyspnoea=True)	P(Dyspnoea=False)
True	0.65	0.35
False	0.30	0.70

True = present, False=absent